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A Deep Learning-Based Approach for Precise Emotion Recognition in Domestic Animals Using EfficientNetB5 Architecture

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Abstract: The perception of animal emotions is key to enhancing veterinary practice, human–animal interactions, and protecting domesticated species’ welfare. This study presents a unique emotion classification deep learning-based approach for pet animals. The actual and emotional status of dogs and cats have been classified using a modified EfficientNetB5 model. Utilizing a dataset of images classified into four different emotion categories—angry, sad, happy, and neutral—the model incorporates sophisticated feature extraction methods, such as Dense Residual Blocks and Squeeze-and-Excitation (SE) blocks, to improve the focus on important emotional indicators. The basis of the second strategy is EfficientNetB5, which is known for providing an optimal balance in terms of accuracy and processing capabilities. The model exhibited robust generalization abilities for the subtle identification of emotional states, achieving 98.2% accuracy in training and 91.24% during validation on a separate dataset. These encouraging outcomes support the model’s promise for real-time emotion detection applications and demonstrate its adaptability for wider application in ongoing pet monitoring systems. The dataset will be enlarged, model performance will be enhanced for more species, and real-time capabilities will be developed for real-world implementation.

Keywords: emotion detection; domestic animal; anger; happy; sad; EfficientNetB5



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1. Introduction

Accurately recognizing and understanding domestic animal emotional states is crucial for advancing veterinary practice, improving human–animal bonds, and promoting animal welfare. With the increased prevalence of companion animals, especially dogs and cats, as members of households around the world, there exists the need to develop automated systems to assess their emotions. Correctly identifying these feelings can yield insights crucial for resolving behavioral problems, health issues, or distress, leading to better management and care of our pets. There has never been a greater demand for automated systems that can accurately understand the emotional states of companion animals. Identifying these emotions can help detect behavioral issues, medical issues, or distress, in turn improving pet care and management [1]. For instance, a dog that is “angry” may show a direct stare, bare fangs, and a rigid body posture, whereas a dog that is “happy” may show an open mouth, relaxed ears, and a wagging tail. Conventional image processing methods struggled to capture the visual cues that are the basis of emotion recognition. Existing methods cannot handle complex variations in animal postures, lighting, and occlusions, making advanced machine-learning models an essential alternative [2,3].

Numerous human emotion recognition databases are available and standardized, while animal datasets are often scarce and limited to the species. A big research gap continues to be the lack of diverse datasets that reflect the complex emotional states of animals. Additionally, animal emotions are quite nuanced and differ by species, making this task difficult. These issues require rich novel methods to not only learn hierarchical features but also learn them in a computationally efficient manner. Machine learning techniques, particularly Convolutional Neural Networks (CNNs), have gained significant attention for their effectiveness in recognizing human facial expressions [4]. The study's dataset consists of images of household pets—mostly dogs and cats—classified into four emotional states: neutral, happy, sad, and furious. These categories were established using visible indicators, including posture, body language, and facial expressions. A major obstacle to creating reliable categorization models is the intrinsic complexity of animal emotions combined with species variations. Many earlier studies investigating animal behavior recognition have not sufficiently covered the subtleties of emotion-specific classification in pets. Significant research gaps in this area are further highlighted by the paucity of extensive datasets that reflect the diversity of animal expressions and the sparse use of cutting-edge deep learning algorithms [5]. Deep learning models, particularly Convolutional Neural Networks (CNNs), are excellent at solving complicated visual identification tasks such as emotion classification with high accuracy [6]. Efficient Net has emerged as an appropriate framework to achieve high accuracy while minimizing computational expenses [7,8]. However, typical solutions may not capture the complex details required for animal emotion recognition. To solve this, this study presents a customized EfficientNetB5 architecture that includes Dense Residual Blocks and Squeeze-and-Excitation (SE) blocks. SE blocks recalibrate feature channels to focus on important components of animal emotions, whereas Dense Residual Blocks aid in the learning of hierarchical and complex characteristics. By utilizing these improvements, the model can better manage miniature changes in facial expressions, ear positions, and body language, which are critical for correct emotion classification.

The dataset used in this study includes images of domestic pets, predominantly dogs and cats, in four different emotional states: angry, sad, joyful, and neutral. These categories were identified using observable signs like posture, body language, and facial expressions. Species-specific variations in emotions, postures, and the natural diversity within the same species make it particularly challenging to understand and interpret the feelings of pets [9,10]. Data augmentation techniques such as rotation, flipping, and brightness changes were applied to counter these challenges and improve the dataset's diversity and robustness. Despite all the challenges, the proposed EfficientNetB5 model demonstrated an outstanding accuracy of 98.2% for the training dataset and 91.24% validation accuracy for the validation dataset. In addition, precision, recall, and F1 scores show the model's reliability in real-world applications. Integrating transfer learning with ImageNet pre-trained weights was also critical in tailoring the model to the specific needs of animal emotion recognition.

This study demonstrates the potential of incorporating such models into real-time systems that can track and evaluate pets' mental health in a practical setting. Furthermore, the accuracy gained in this study indicates the proposed model's usefulness and the broader implications of employing deep learning to investigate animal behavior. In future studies, we intend to increase the model's real-time processing capabilities, add more species to the dataset, and investigate how this model may be integrated into pet monitoring systems. Finally, this study contributes to the growing field of animal emotion recognition by providing new resources for improving animal care and fostering a stronger link between owners and their pets. The paper's remaining structure is as follows: Section 2 addresses previous research; Section 3 describes the dataset and data augmentation methods; Section 4

gives the experimental setup, outcomes, and comparative analysis; and Section 5 concludes the study.

2. Related Work

Emotion recognition in domestic animals has garnered increasing attention in recent years, with advancements in deep learning and computer vision techniques playing a pivotal role. Researchers have developed various methods to interpret and classify animal emotions based on facial expressions, vocalizations, and body postures, often employing Convolutional Neural Networks (CNNs) [11] due to their exceptional ability to capture intricate patterns in visual data. CNN-based approaches for emotion recognition have demonstrated success across multiple species. For instance, Corujo et al. [12] developed a combined model of region-based CNN for horse detection and a CNN-based model for emotion prediction and achieved 65% accuracy on test data, demonstrating that it is possible to predict emotion in animals. Deep learning and convolutional neural networks (CNNs) have greatly improved medical imaging by enabling automated analysis and interpretation of intricate visual data in the medical field [13,14]. Their capacity to learn hierarchical features has proven essential for tasks like disease classification, detection, and segmentation, yielding high accuracy levels on par with expert performance [15–17]. This study highlights the potential of CNNs to accurately classify canine emotions and open new avenues for enhancing human–animal interactions. The authors of [18] created an automated cat pain recognition system, utilizing facial image analysis with ResNet50 models [19], achieving accuracy above 72%. Their work underscores the application of CNNs in non-human species for objective emotion assessment.

Further advancements in CNNs for animal emotion recognition have been made by [20], who proposed a CNN-based classification model for dog emotions, such as sadness, happiness, relaxation, and anger. Optimized via Stochastic Gradient Descent, this model achieved a validation accuracy of 91.24%, demonstrating CNNs' superior performance in the field compared to traditional models like Support Vector Machines (SVMs) and k-nearest Neighbors (KNNs). The authors of ref. [21] also introduced a real-time system to recognize domestic cat emotions, categorizing expressions into "Pleased," "Angry," "Alarmed," and "Calm" using CNNs. This system's robustness and high accuracy show the potential to improve communication between humans and pets and enhance veterinary care. Moreover, other studies have utilized pose estimation and key point detection for animal emotion recognition. The authors of [22] applied DeepLabCut for posture analysis to recognize emotions such as anger, fear, happiness, and relaxation in dogs, achieving an emotion classification accuracy of 60–70%, surpassing human capabilities in detecting canine emotions. This method demonstrates how machine learning models can automate emotion recognition through visual analysis, highlighting the growing potential of deep learning in understanding animal behavior. Table 1 offers a thorough summary of the methods and datasets currently used in pet emotion recognition. The approaches and data sources that have influenced developments in this discipline are highlighted in this synopsis.

In recent years, applying pre-trained models like Efficient Net [23,24] has gained traction for emotion recognition tasks. The authors leveraged EfficientNetB0 with transfer learning for dog emotion recognition, achieving a classification accuracy of 78% [25]. The success of EfficientNetB5 demonstrates its effectiveness in recognizing subtle emotional cues in animal images, providing a foundation for further developments in animal welfare, behavioral analysis, and veterinary care. Additionally, explored the potential of deep learning models for predicting canine emotions, including VGG16, InceptionV3, and Vision Transformer [26]. Their study demonstrated that EfficientNetB0 outperformed

other models, achieving an accuracy of 80.5% in recognizing emotions such as anger, fear, happiness, and relaxation. This underscores the growing interest in advanced deep learning architectures like Efficient Net for animal emotion prediction. In computer vision, CNNs and deep learning have greatly improved image classification by enabling reliable feature extraction and attaining high accuracy for object recognition and illness detection tasks. Research employing topologies like ResNet, VGGNet, and EfficientNet has established standards for precisely categorizing intricate visual data [27–31]. These advancements illustrate the increasing sophistication of emotion recognition systems for domestic animals, driven by applying state-of-the-art deep learning techniques. While much progress has been made, the field still faces challenges, particularly in enhancing model accuracy and generalizing across different species. Our work builds upon these efforts, proposing a customized EfficientNetB5 architecture to further improve domestic animal emotion recognition [32,33]. Focusing on deep learning techniques that integrate transfer learning and fine-tuning, this study aims to contribute to more accurate and reliable emotion recognition systems that can be applied in various animal welfare and veterinary contexts.

Table 1. Summary of various existing classification techniques.

Authors (Year) [Reference]	Animal Type and Emotions	Feature Extraction	Dataset	Evaluation Metric	Performance (%)
Corujo et al. (2021) [12]	Horse emotions: Alarmed, Annoyed, Curious, Relaxed	CNN	Self-generated	Accuracy, AUC	65
Feighelstein et al. (2022) [18]	Cat data: Pain, No pain	ResNet50	The dataset used in the paper by Finka et al. [34]	Accuracy, Precision, Recall	72
Tanwar (2024) [20]	Dog emotions: Angry, Happy, Relaxed, Sad	CNN	Dog emotions Dataset from Kaggle	Accuracy	97.6
Ali et al. (2024) [21]	Cat emotions: Pleased, Angry, Alarmed, Calm	CNN	Self-generated	Accuracy, Precision, Recall, Confusion matrix, and F1 Score.	97.9
Ferres et al. (2022) [22]	Dog emotions: Anger, Fear, Happiness, Relaxation	DeepLabCut	Self-generated	Accuracy	60–70
Xie (2024) [25]	Dog emotions: Sad, Happy, Angry, Relaxed	EfficientNetB0	Dog emotions Prediction from Kaggle	Accuracy, AUC, Precision	78
Kakani et al. (2024) [26]	Dog emotions: Anger, Fear, Happiness, Relaxation	VGG16, EfficientNetB0, and InceptionV3	Self-generated	Accuracy, Precision, Recall, F1 Score	80.5

3. Materials and Methods

Understanding domestic animals' behavior and enhancing their welfare depend heavily on their ability to recognize emotions. In this study, we provide a customized EfficientNetB5 model to categorize companion animals into four emotional states: angry, sad, joyful,

and neutral. The model uses deep learning to provide precise real-time emotion recognition for useful applications in animal care, as shown in the flow diagram in Figure 1.

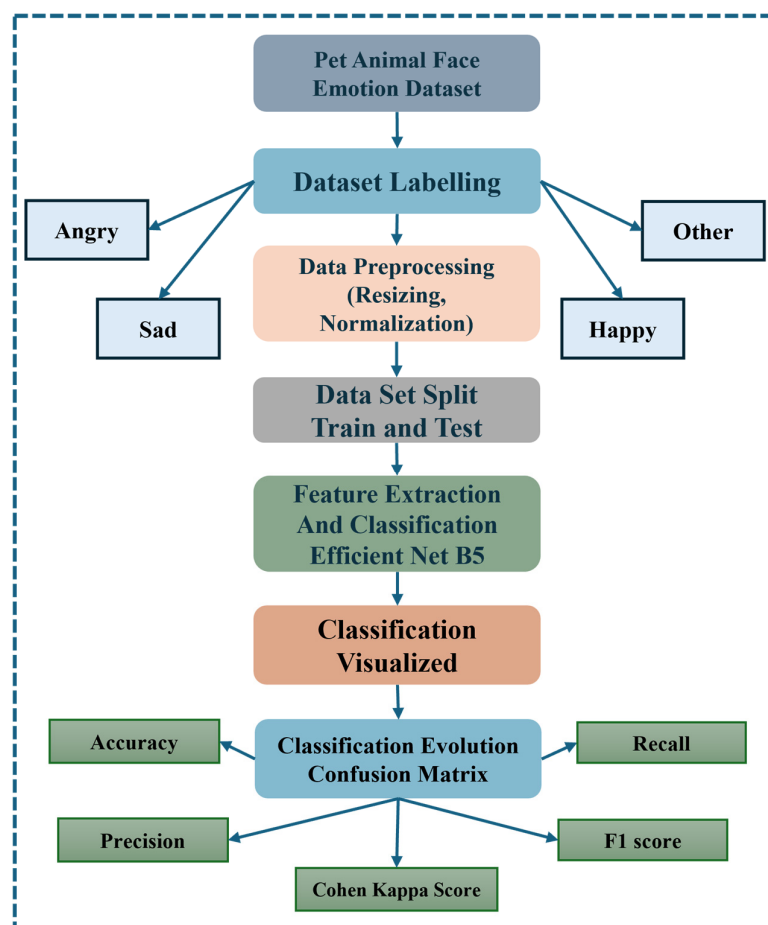


Figure 1. Flow diagram of pet animal face emotion detection.

Data Collection: Our experiment used a dataset of domestic pet animal emotion photographs from the Kaggle platform (<https://www.kaggle.com/datasets/anshtanwar/pets-facial-expression-dataset>, accessed on 13 November 2024). The dataset includes standard domestic pets in various emotional states, namely dogs and cats. To help with accurate emotion classification, we focused on four distinct emotion categories: angry, sad, joyous, and neutral/other. The images were manually categorized using facial expressions, body language, and other environmental cues that indicate emotional states. Carefully analyzing each image to ensure it belonged to the right emotional group increased the dataset's reliability. The dataset was separated into four primary categories to reflect the different emotional states: images in the happy class depict pets lounging with signs of relaxation, such as wagging tails or relaxed ears; images in the sad class depict animals with downcast eyes, slumped postures, or other signs of low energy or distress; and, finally, images in the neutral/other category depict unclear or neutral emotional states, such as a calm and non-expressive postures. Images in the angry class depict tense or aggressive postures, frequently with bare teeth or furrowed brows. Figure 2 displays representative examples of each emotional type. The first row shows images in the angry state, while the second row displays images in the neutral or otherwise state. The dataset was split into three subsets to guarantee a fair assessment of the model's performance. Two hundred photos were set aside for validation: 120 for testing and 2000 for training. This separation makes it possible to evaluate the model's precision, resilience, and ability to generalize new data in detail.



Figure 2. Categorical images that show the four emotional states of domestic animals. Images in the first row are categorized as Angry, in the second row as Neutral/Other, in the third row as Sad, and in the fourth row as Happy.

Data Preprocessing: Noise, blurriness, and unnecessary images in the dataset must be removed to prevent the models' results from deteriorating. When we used test images that had never been seen before, we evaluated the models' ability to generalize by ensuring they had been properly trained and tested by cleaning them up. The noise and blur removal phase aims to identify and eliminate noisy or fuzzy photos from the dataset.

Data Augmentation: rotation, cropping, and noise introduction techniques create new data samples and increase the dataset's diversity.

This preprocessing phase aims to enhance data quality by minimizing visual artifacts that can impair model performance. The noise was suppressed using Gaussian and median filtering, and blur images were made clearer using unsharp masking. The dataset did not include low-resolution photos that could not be sufficiently adjusted, severe noise, or distortion. These improvements increased the clarity and general quality of the input images, guaranteeing more precise feature extraction of the body postures and facial expressions essential for emotion recognition, illustrated in Figure 3. Ultimately, this step improved the proposed model's robustness and dependability.

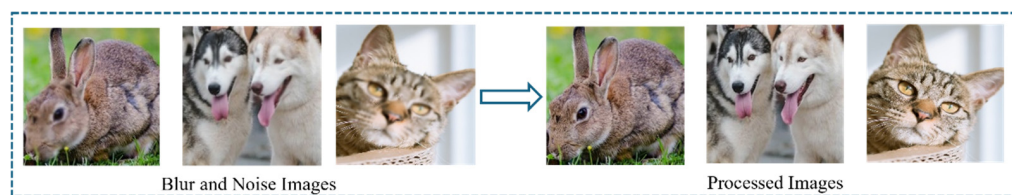


Figure 3. The left panel displays the original images, while the right panel displays the corresponding pre-processed images after applying the noise and blur reduction algorithms.

Class Balance: In classification tasks, there must be enough samples for each class to prevent class imbalance, which could skew the model's performance.

Data Splitting: To precisely evaluate the model's performance and prevent data leaks, the dataset is divided into training, validation, and test sets. These data-cleaning techniques enable the preparation of a robust and reliable dataset, which is essential for the training and testing of machine learning models.

Removal of Duplicates: Identifying and removing duplicate photos from the dataset is crucial to avoid bias and redundancy and guarantee that the model learns from unique examples.

Methodology: The foundation of the proposed model architecture for emotion recognition in domestic animals is the EfficientNetB5 model, which is selected for its efficient balance between computational cost and classification performance on visual recognition tasks. Leveraging its pre-trained weights from ImageNet at initialization, the EfficientNetB5 backbone enables the model to transfer essential, general-purpose features while focusing upper layers on the specific requirements of emotion classification in animals. The input layer is configured to process RGB images resized to 224×224 pixels, defining an input shape of $(224, 224, 3)$, which maintains consistency across the samples, shown in Figure 4. To enhance the model's capacity for extracting high-dimensional features, the EfficientNetB5 is modified to exclude its final classification layer (including top = False), allowing for custom task-specific layers to be integrated later in the architecture. During training, all layers except the last 20 are frozen to retain pre-trained weights, which preserves foundational image features while granting the model flexibility in fine-tuning upper layers for more nuanced aspects of emotion recognition.

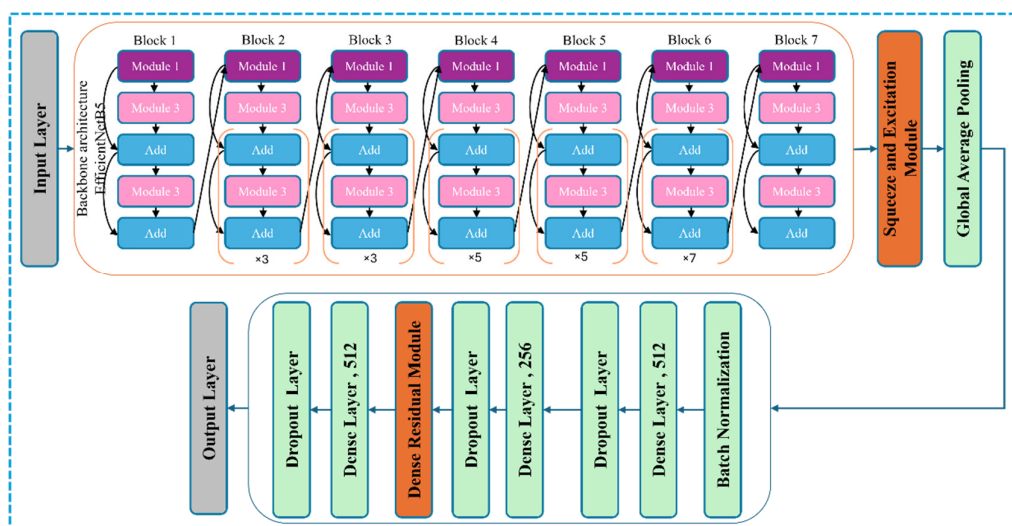


Figure 4. Backbone architecture of EfficientNetB5 model for pet emotion classification.

An iterative process that combined grid search and manual tuning was used to determine the hyperparameters for the proposed Modified EfficientNetB5 model. Values between 0.00001 and 0.01 were tested to maximize the learning rate; 0.0001 was chosen as

the ideal value based on validation performance. Similarly, batch sizes of 16, 32, and 64 were assessed; however, a batch size of 32 was selected since it offered the optimum trade-off between computing efficiency and training stability. To reduce overfitting, dropout rates were examined between 0.2 and 0.5; the best reliable findings were obtained at 0.3.

The model's output layer is a fully connected dense layer equipped with softmax activation. This layer classifies input images into one of four predefined emotion categories: angry, sad, happy, and neutral. The softmax function converts the final layer's logits into a probability distribution across these classes, allowing the network to output predictions with interpretable probability values. This setup ensures that the model generates informed, probabilistic predictions, maximizing interpretability and providing confidence in the classification of each image. The Adam optimizer is employed for optimization, which adapts learning rates based on the first and second moments of gradients to achieve a balanced and stable convergence. A learning rate of 0.0001 is specified, maintaining stability and preventing abrupt shifts in weight updates. As the task is multi-class, categorical cross-entropy is selected as the loss function, measuring the dissimilarity between the predicted and true class distributions.

Squeeze-and-Excitation (SE) Module Integration:

A Squeeze-and-Excitation (SE) block is added immediately after the input passes through the EfficientNetB5 backbone to refine the model's focus on the most informative feature channels, shown in Figure 5. The SE module enhances attention by adaptively recalibrating channel-wise feature responses, thereby prioritizing channels that convey critical information about animal emotions. Specifically, the SE block applies a Global Average Pooling (GAP) operation across the spatial dimensions of the input tensor X , which has dimensions $H \times W \times C$, producing a compact $1 \times 1 \times C$ representation that encapsulates each channel's global spatial context. This representation then passes through a fully connected Dense layer to model channel dependencies; typically, a reduction ratio of 16 is applied to compress channel information, enabling the network to learn an intermediate representation.

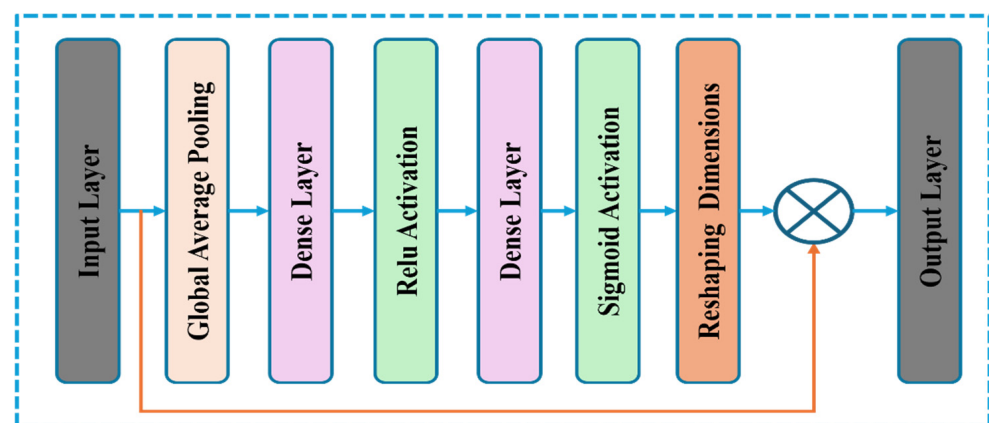


Figure 5. Squeeze-and-Excitation module architecture.

ReLU activation is employed here to introduce non-linearity, after which a second Dense layer restores the tensor to the original channel dimension C . A sigmoid activation function generates channel-wise attention weights in the range $[0, 1]$, which are reshaped to $1 \times 1 \times C$ and broadcast across spatial dimensions H and W . The recalibrated attention map is then applied to the input tensor through element-wise multiplication, producing a refined output that highlights salient channels while attenuating less relevant ones. By guiding the model's focus on the most critical features, the SE module enhances performance on

tasks with complex feature discrimination requirements, such as emotion identification in animals.

Mathematically, the Squeeze-and-Excitation block is represented as follows:

$$\tilde{X}_c = \sigma \left(W_2 \text{ReLU} \left(W_1 \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i,j) \right) \right) \cdot X_c \quad (1)$$

where X_c is the input feature map for the channel, C , W_1 , and W_2 are the weight matrices for the fully connected layers, σ represents the ReLU sigmoid activation function, H and W are the height and width of the input feature map, and $\tilde{X}_c$ is the recalibrated output for the channel C .

Dense Residual Block:

A custom Dense Residual Block is incorporated further to extend the model's depth and representation capabilities, as shown in Figure 6. This block is structured to enhance feature learning by integrating residual connections within a densely connected framework. Initially, the input tensor X is processed by a Dense layer with a set number of units. ReLU activation is applied to each unit, allowing the model to capture non-linear relationships within features. An optional kernel regularization is employed here to control overfitting. The Dense layer is followed by Batch Normalization, which stabilizes training by standardizing the output activations. If configured, a dropout layer follows Batch Normalization to enhance regularization by randomly omitting neurons.

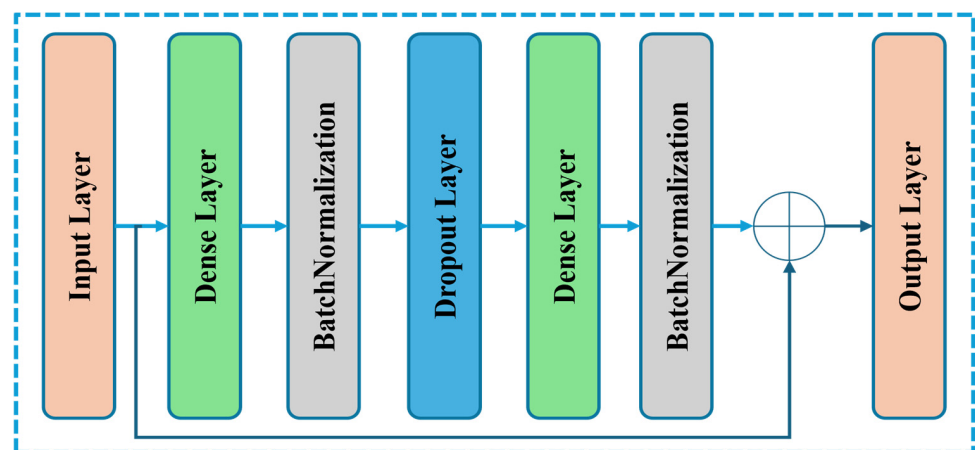


Figure 6. Dense Residual Block architecture.

A second Dense layer with an identical configuration processes the output of this first Dense–Batch Normalization pair. To preserve critical information, a residual connection adds the original tensor X back to the production of the second Dense–Batch Normalization pair. This shortcut connection alleviates the vanishing gradient problem and allows for efficient gradient flow throughout the network, thereby improving the learning of deeper layers. A final ReLU activation is applied to the combined output to introduce non-linearity, ensuring that the Dense Residual Block effectively captures complex relationships in the data. This architecture is especially suited for emotion classification, where subtle feature differences are prevalent and require precise representation.

Mathematically, the Dense Residual Block is represented as follows:

$$Y = X + \text{BN}(W_2 \cdot \phi(\text{Dropout}(\text{BN}(W_1 \cdot X + b_1), p)) + b_2) \quad (2)$$

where X is the input to the block, W_1 and W_2 are the weight matrices of the Dense layers, b_1 and b_2 are the biases for the Dense layers, $\text{BN}(\cdot)$ denotes the Batch Normalization

function, Dropout p) is the dropout function with dropout rate p , and $\phi(\cdot)$ denotes an activation function (e.g., ReLU).

4. Results

The model training time, along with the hardware and software requirements, is there to improve the assessment of computing efficiency. A system with an Intel core i7 processor, 32 GB of RAM, and an NVIDIA RTX 4080-Ti GPU (24 GB VRAM) was used for the research. TensorFlow 2.9.0 and Python 3.10 were part of the Windows operating system. A maximum of 150 epochs were used to train the Modified EfficientNetB5 model, with early stopping implemented in response to validation loss. Each training run took about three hours on average, indicating the proposed method's computational effectiveness and applicability for emotion recognition tasks.

The proposed Modified EfficientNetB5 model performs better than alternative deep-learning models regarding pet animal face detection. Achieving a training accuracy of 98.20% and testing accuracy of 91.24%, the proposed model outperformed widely used architectures like MobileNet, egg-16, Inception V3, DenseNet, Xception Net, and Resnet-50. Along with the highest Cohen Kappa Score (Ckp) of 91.88%, it also recorded exceptional performance measures, such as a precision of 90.98%, recall of 92.14%, and F1 score of 91.20%, illustrated in Table 2. We used confusion matrices in conjunction with a set of related metric parameters, including accuracy (Acc), precision (Pre), recall (Rec), F score (F-Score), and Cohen Kappa Score (Ckp), to quantify the performance of our model during the experiment of this work. The true positive (TP), false positive (FP), true negative (TN), and false negative (FN) parameters that were obtained from the confusion matrix were used to construct these measurements. The following formulas were used to compute the confusion metrics:

Table 2. Performance comparison of various classification techniques with proposed pet animal face emotion detection models.

Deep Learning Model	Training Accuracy (%)	Testing Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Ckp Score (%)
Mobile Net	68.52	52.12	51.07	50.53	51.90	52.51
VGG-16	84.52	73.52	74.32	73.80	74.47	74.21
Inception V3	48.93	41.23	42.10	42.25	42.14	43.14
Alex Net	67.70	51.40	52.03	51.00	52.50	51.99
Exception	48.27	43.99	44.25	43.97	44.09	44.52
Dense Net	67.99	58.30	58.30	57.10	56.99	59.30
Res Net-50	91.95	89.10	88.99	89.00	89.08	89.30
Proposed Modified EfficientNetB5	98.2	91.24	90.98	92.14	91.20	91.88

The accuracy of a model is determined by how well it can categorize examples. It is the ratio of all observations to correctly predicted observations (the sum of true positives and true negatives). The mathematical expression for accuracy is

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3)$$

Positive forecast accuracy is measured using a metric called precision. It is defined as the ratio of positive estimations to positive predictions made by the model. In mathematics, precision is shown as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4)$$

Recall is a statistic used to evaluate a model's ability to identify positive cases. It is the ratio of true positive expectations to actual positive cases. The following is a mathematical expression for recall:

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (5)$$

F1-Scr assesses the rates of correctly predicted genuine positive samples, which is also known as sensitivity and true positive rate.

$$\text{F1} - \text{Scr} = \frac{2 * (\text{Rec} * \text{Pre})}{(\text{Rec} + \text{Pre})} \quad (6)$$

The Cohen Kappa Score (Ckp), a statistical measure of inter-rater agreement for categorical categories is defined as

$$\text{Ckp} = \frac{C_0 - C_e}{1 - C_e} \quad (7)$$

C_0 is a high percentage of observed agreement between raters and classifiers.

C_e is the hypothetical probability of chance agreement.

Table 2 compares parameters, including training and testing accuracy, precision, recall, F1 score, and Cohen's Kappa score, to show how well various deep learning models perform at detecting facial emotions in pets. The best testing accuracy of 91.24% is demonstrated by the suggested Modified EfficientNetB5 model, which also has a precision of 90.98%, recall of 92.14%, F1 score of 91.20%, and Cohen's Kappa score of 91.88%. When identifying tiny emotional cues in pet faces, these measurements show a good mix of accuracy and dependability. With an F1 score of 89.08% and a testing accuracy of 89.10%, ResNet-50 also demonstrates competitive baseline performance. On the other hand, models like VGG-16, MobileNet, and DenseNet perform quite well, with testing accuracies ranging from 52.12% to 73.52%. Lower performance from Inception V3 and Xception suggests difficulty identifying pet emotions. These results demonstrate the Modified EfficientNetB5 model's benefit for precise and reliable emotion identification, highlighting its potential for practical use in pet emotion recognition systems.

The training and validation accuracy of several deep learning models for pet emotion identification is shown in Figure 7. With steady accuracy across training and validation sets, the Modified EfficientNetB5 model exhibits the best and most reliable performance, with great generalization and little overfitting. While still achieving comparable results, ResNet-50's validation accuracy is marginally lower than EfficientNetB5's. Comparatively, VGG-16 demonstrates a small improvement in accuracy but notable validation variations, indicating difficulties in accurately identifying emotional patterns. These outcomes highlight the Modified EfficientNetB5 model's exceptional performance, making it the best model for precise and trustworthy pet emotion recognition.

Confusion matrices evaluating the effectiveness of various deep-learning models for pet emotion classification are shown in Figure 8. With the best accuracy and clearly defined clusters along the diagonal, the Modified EfficientNetB5 model successfully identifies most emotional categories with little misclassification. Although it performs marginally worse than EfficientNetB5 in differentiating between similar emotions like "happy" and "sad", the ResNet-50 model likewise does well, displaying strong clustering with fewer mistakes.

Conversely, models like DenseNet and VGG-16 show more off-diagonal entries, indicating difficulties in distinguishing between emotions. These findings demonstrate the Modified EfficientNetB5 model's superiority over inaccurate emotion categorization, which positions it as a viable option for real-world pet emotion detection applications.

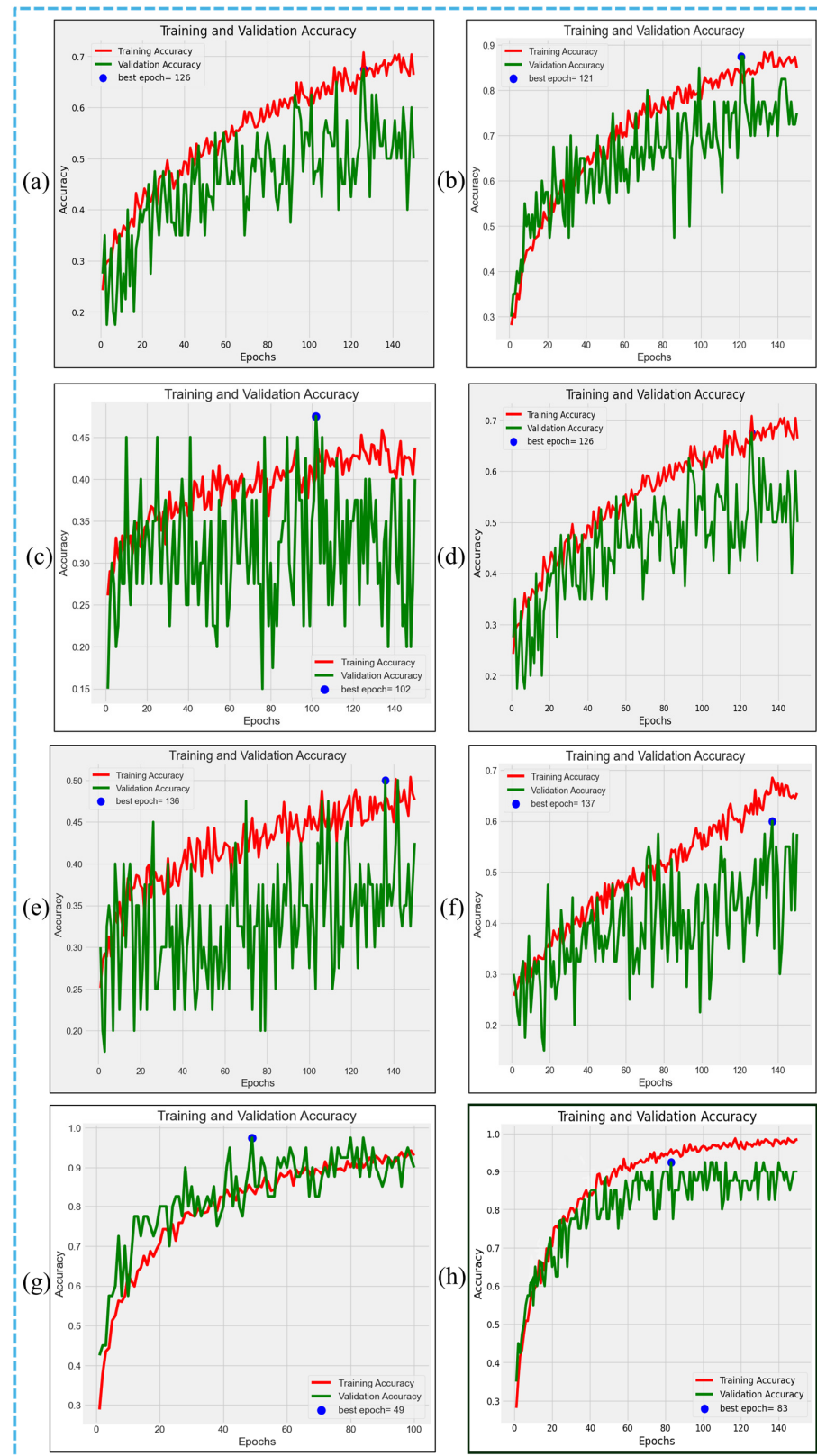


Figure 7. Comparison training and testing accuracy of (a) Mobile Net, (b) VGG-16, (c) Inception V3, (d) Alex Net, (e) Exception, (f) Dense Net, (g) Res Net-50, and (h) Proposed Modified EfficientNetB5.

Angry	28	4	6	2	Angry	34	2	3	1	Angry	13	2	19	6	Angry	16	9	2	13
Other	1	26	4	1	Other	3	29	0	0	Other	1	8	4	0	Other	1	7	10	11
Sad	1	3	33	8	Sad	6	7	29	3	Sad	2	0	41	2	Sad	7	13	11	14
Happy	3	2	6	32	Happy	3	1	8	31	Happy	6	0	22	15	Happy	9	7	8	19
Angry	(a)				Angry	(b)				Angry	(c)				Angry	(d)			
Other					Other					Other					Other				
Sad					Sad					Sad					Sad				
Happy					Happy					Happy					Happy				
Angry	38	1	1	1	Angry	17	9	5	9	Angry	36	4	6	2	Angry	40	0	0	0
Other	1	28	2	1	Other	3	21	4	4	Other	1	26	4	1	Other	0	32	0	0
Sad	6	0	36	3	Sad	9	9	20	7	Sad	1	1	33	8	Sad	3	0	41	1
Happy	4	0	4	35	Happy	5	9	14	15	Happy	3	0	6	32	Happy	0	2	1	40
Angry	(e)				Angry	(f)				Angry	(g)				Angry	(h)			
Other					Other					Other					Other				
Sad					Sad					Sad					Sad				
Happy					Happy					Happy					Happy				

Figure 8. Comparison confusion matrices of (a) Mobile Net, (b) VGG-16, (c) Inception V3, (d) Alex Net, (e) Exception, (f) Dense Net, (g) Res Net-50, and (h) Proposed Modified EfficientNetB5.

Sample prediction results for the suggested Modified EfficientNetB5 model for pet facial emotion categorization are shown in Figure 9. The figure shows both instances where the model misclassified the emotions and correctly classified photos, where the model predictions align with the actual emotional labels. The model is generally very accurate, recognizing emotions like “happy,” “angry,” “sad,” and “other” the majority of the time. A few photos show misclassifications, where minute posture and facial expression variations may make predictions more difficult. These findings illustrate the model’s resilience and efficacy in identifying animal emotions while pointing out areas requiring more development in intricate situations.

This work uses a modified EfficientNetB5 model that has been improved with Squeeze-and-Excitation (SE) blocks, regularization-enhanced Dense layers, and a bespoke residual block to provide a novel method for emotion recognition in domestic animals. Because assessing animals’ delicate and varied expressions presents unique problems, the suggested model seeks to fill a vacuum in emotion recognition research, which has primarily concentrated on humans. A model that can distinguish fine-grained traits while retaining computing efficiency is necessary because animal emotions, in contrast to human emotions, are expressed through more subtle facial and postural cues. The suggested model outperforms traditional architectures like MobileNet, VGG-16, Inception V3, and ResNet-50 in identifying complex emotional states across a variety of species, according to results from accuracy and confusion matrix analyses. In particular, a consistent learning curve and good performance on unknown data demonstrated that the updated EfficientNetB5 model decreased overfitting and increased validation accuracy. The adaptive channel recalibration of the SE block, which enables the network to prioritize pertinent features connected to every emotion category, is responsible for this improvement. The SE block efficiently directs the model’s attention by giving higher weights to channels that convey important emotional information. This improves classification accuracy, especially when traditional models had trouble distinguishing between similar emotions like “sad” and “neutral.” Practically speaking, the findings of this study offer promising prospects for developing real-time systems for monitoring animal emotions, which have the potential to transform fields including animal welfare, veterinary care, and behavior research. The proposed model’s high accuracy and efficiency make it suitable for integration into mobile

and embedded systems, enabling broad deployment in scenarios that call for continuous and non-intrusive.

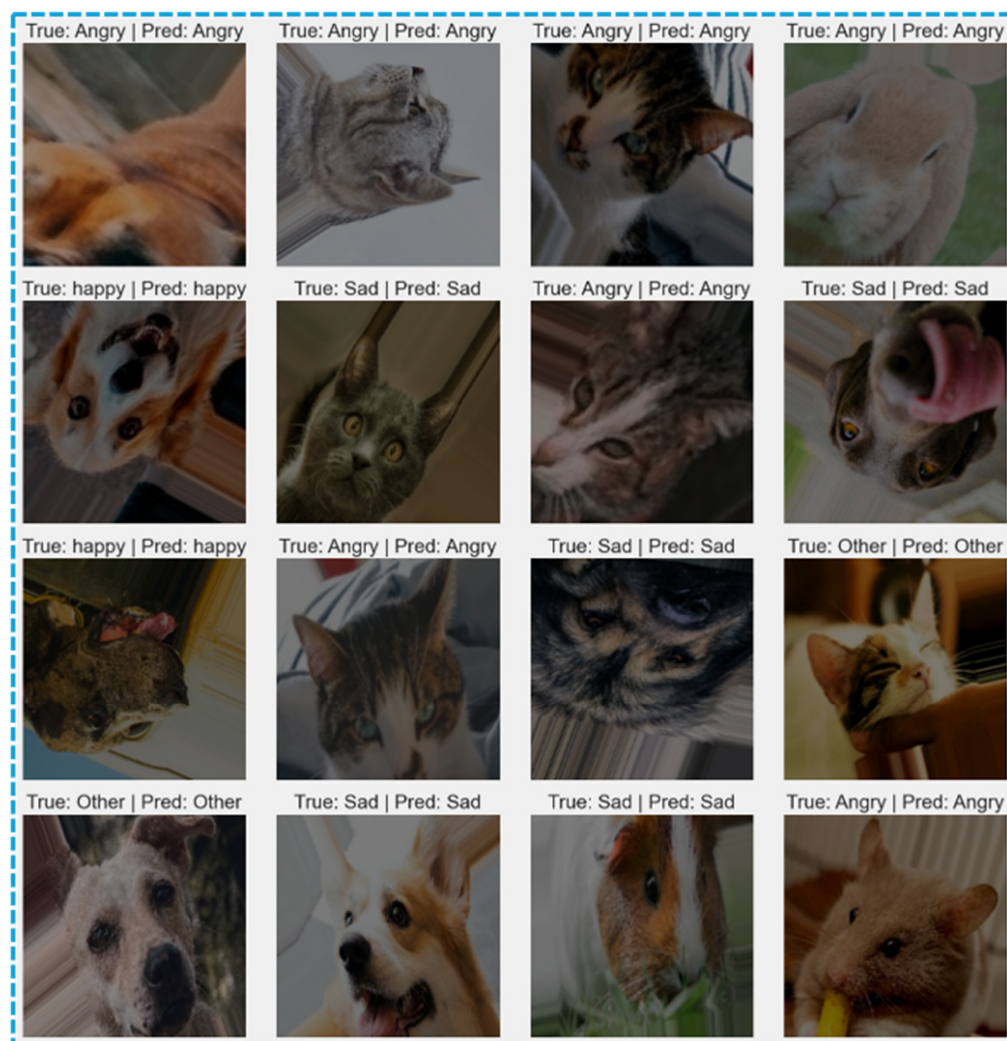


Figure 9. Prediction results of proposed Modified EfficientNetB5 model.

5. Conclusions

This study describes a deep learning method that uses an EfficientNetB5 model tweaked and fine-tuned with residual Dense blocks to enhance classification performance and detect emotions in household pet animals. The model's close alignment between training and validation performance indicates that it underwent minimal overfitting, achieving a high overall accuracy of 89% on a dataset comprising four unique emotional states: angry, sad, joyful, and neutral (other). According to the confusion matrix and classification report, the model does especially well in identifying the angry, cheerful, and neutral classes, achieving F1 scores of 0.88, 0.89, and 0.96, respectively. Despite this, the model's overall performance remained balanced, demonstrating its great potential for use in pet emotion recognition systems in the real world. According to the findings, animal complex emotional states can be accurately classified from picture data using deep learning and a reliable pre-trained model such as EfficientNetB5. Future research could concentrate on expanding the dataset, particularly for the difficult sad category, and adding more image augmentation methods to improve the model's functionality even more. This research offers a solid foundation for developing automated emotion identification systems to improve animal welfare and promote deeper human–animal connections.

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